

The Indus seal text: an established frame around an unread individual designation. Calibrated tests against Markov nulls, with the Umma sealing as an exported standard formula

Draft, 3 October 2026 (revised after the red-team audit S-DARK-41, the Markov re-grading S-DARK-54 and the follow-up loops S-DARK-43, 45, 48, 52, 53, 56, 58 and 59; the 'credential grades' claim of the 2 October draft is withdrawn and the uniqueness-based 'not names' contrast is retired, see §3.1–3.2). Authors: P. Garg (and co-authors to be agreed). The code and data for every number are in this repository; the strategy IDs (S###, S-DARK-##) point to the log in STRATEGIES.md. Unless stated, counts come from data/derived/merged-corpus-canonical.json, built from an older inscriptions.csv (5,369 rows; S-DARK-23); the rebuilt corpus (5,533 texts) moves no result (S-DARK-41.1).

Abstract

Most attempts to decipher the Indus script assume that seal inscriptions record owners' names, spelled phonetically. Mukhopadhyay (2023) argued instead that the inscriptions contain no proper nouns and served licensing, taxation and access control. We test the structure of the inscriptions on the merged Wells/ICIT corpus (3,599 distinct texts), with Mahadevan's 1977 transcription (IM77) as a transcription check, against first- and second-order Markov chains fitted within site and object type, and on held-out sites. (1) **An established frame.** A seal text opens with one of five opener units (W817/W861/W820 + the marker W2; W920/W692 + W60), closes with one of a closed set of mutually exclusive closers, and may take a tablet suffix. Against a fitted bigram chain the frame survives where its parts are not adjacency facts: the marker W2 occurs once per text (3 texts with two W2 against 34 expected under the chain; 3.5 against 47 per 1,000 texts), 9 closers are mutually exclusive against a null maximum of 8, closer mutual exclusion is 0.042 against 0.164, the two-closer ban 0.023 against 0.069, the ban on repeating a sign 0.038 against 0.122, and the non-adjacent fixed order of signs 0.48 against 0.16 (held-out sites 0.65 against 0.03). The closer-last rule holds on held-out sites at 0.42 against a shuffled maximum of 0.21 (S-DARK-41, S-DARK-54). The frame is at full strength in the earliest dated phase (opener-first 0.26 against 0.24 overall, closer-last 0.68 against 0.67, order agreement 0.97); growth of more than about 10% is excluded, and the count-and-unit layer is older than the seal frame (S-DARK-43). On seals the two big cities are one population (every divergence component within $\times 0.96$ –1.20 of a within-city split, none significant), the small sites sit inside that band on 31–32 of 32 tests, and the between-city excess is smaller than in Ur III legends or Linear B (S-DARK-48.1, 48.3). (2) **The credential-grade claim is withdrawn.** In the 2 October draft, nesting of short texts inside longer ones ($4.1\times$) and reuse of big-city runs at held-out sites ($5\times$) were read as grades of a credential. A first-order Markov chain fitted to the same texts reproduces the nesting exactly (0.152 against 0.152) and the run reuse ($1.3\times$; $0.7\times$ against a big-city chain). Both were artefacts of a within-text shuffle null (S-DARK-41). The credential model is now grade C (S-DARK-54). (3) **The designation slot.** Between opener and closer stands a 2–3-sign element that is unique to its seal. It carries 69% of a seal's information (18.7 of 27 bits, at 7.1–7.5 bits per sign) and is 8% predictable from its own statistics, against 41% for the frame; through the same code an Ur III legend puts 65–71% of its bits in the name (S-DARK-59). One-sign seals keep this element, not the closer (86% against 7%; S-DARK-58). Its elements form one pan-Indus pool, blind to office and city (90% of elements under two or more closers, 97% at two or more sites; S-DARK-45). We call it an individual designation written logographically. Its relation to personal names is open: the

uniqueness contrast with Ur III names is retired (one-per-person name lists match Indus middles to three decimals), but no name list lacks a closed edge set, which Indus middles do (top-10 initial and final elements cover 0.24 and 0.27 of middles, the random-string value, against 0.29-0.63 for Ur III, Old Babylonian, Linear B and Latin names; S-DARK-56). A comparison with logographically written names (loop 63) is pending. (4) **Inventory.** After every defensible merge about 670 signs are in use, about 900 in all (Chao1 913, 95% CI 830-997); a syllabary of 60-120 signs is excluded at every merge level (S-DARK-52). (5) **Nameless quantity seals.** Thirty-one seals from eight sites carry only 'opener · W2 · N · TREE', with $N = 3-6$, the same counts as the Harappa voucher tablets. The Indus sealing reported from Umma carries this formula, '... W2 · 4 · TREE', which is the complete text of seven home seals (S335-S346). (6) **Seal-sealing flow.** All 18 clay sealings that match a seal text were found at a different site from that seal, mostly Mohenjo-daro → Lothal; a site-permutation null predicts most of this (S324-S330). (7) **Pots and other objects.** Pots carry one-sign labels from the same symbol set; no object class carries running text (mean 3.3 signs over 5,369 objects; maximum 17; S354-S355). The frame can be read by function; no sound value has been recovered, and the statistical key-fitting that claims sound values fails its planted controls (S107, S113-S137, S367). We keep the testable anchor of the tree sign as a counted unit (grade B) and timber (grade C), since timber (gesh ab-ba Meluhha) is the Meluhhan good most often recorded at Umma.

1. Background

- **Structure.** Indus seal texts have fixed beginners and enders, a fixed direction and correlated sign order (Yadav et al. 2009; Rao et al. 2009). Long texts divide into recurring segments (Yadav, Vahia, Mahadevan & Joglekar, 'Segmentation of Indus texts').
- **Function.** Mukhopadhyay (2023, *Humanities and Social Sciences Communications*) argued, largely from archaeological context and qualitative script-internal evidence, that the inscriptions encode no proper nouns and served taxation, trade and craft licensing and access control. Earlier readings (Mahadevan, Parpola) took the texts to be names plus titles.
- **Decipherment by statistics.** Statistical decipherment can produce convincing but wrong readings: planted keys 'read' the same symbols as English, Sanskrit or Tamil (arXiv 2608.02999). Our own planted-language controls agree (S107-S137).
- **What is missing, and what this paper adds.** Published structural claims about the Indus text have been tested against within-text shuffles, a null that any sign-to-sign chain beats. This paper tests the frame against fitted Markov chains and on held-out sites, reports which claims survive, and measures how much of a text the surviving rules explain.

2. Data

- **Main corpus.** The merged Wells/ICIT corpus (data/derived/merged-corpus-canonical.json): 5,369 inscribed objects, 3,599 distinct texts after deduplication by site and text, with object type, emblem, find-spot and dimensions (data/raw/inscriptions.csv). Allograph merges are soft and every test was run at three levels (seq_raw, seq_strong, seq_all; S268); no result below depends on a merge level unless stated.
- **Transcription check.** IM77 (Mahadevan 1977), used for transcription-robustness checks through a Wells-Mahadevan bridge of 142 established and 129 proposed entries (97% of entries hold outside the training cities; S-DARK-27, S-DARK-41). About 70% of its texts are the same objects as in the main corpus (S312), so agreement with IM77 is not independent replication. The 324 IM77-only texts are new material.

- **Held-out sites.** Texts from sites other than Mohenjo-daro and Harappa (424 complete direction-recorded texts, 890 at the loosest filter), and the 239 texts from sites absent from IM77 (S349).
- **Null models.** 200 first-order and 200 second-order Markov chains fitted within site × object type with lengths kept, an END-state chain for finality claims, and the S366 generator (S-DARK-54).
- **Comparison corpora.** Mesopotamian seal legends from CDLI (ATF export; 6,240 names, 1,348 distinct Ur III seal-owner names), Old Babylonian seal names, Linear B personnel series (DAMOS), Latin EDH names, Proto-Elamite entries (S-DARK-56, S-DARK-59).
- **Meluhha attestations.** The CDLI Meluhha attestations (214 texts).

3. Results

3.1 The frame: what survives a fitted chain (S289-S303, S-DARK-41, S-DARK-54)

Model. A seal text consists of an opener unit (W817, W861 or W820 followed by the marker W2; W920 or W692 followed by W60; S331), a middle (§3.2), a closer from a closed paradigm (the jar W740 as default, and 806-154/158, 550/555-527, 220/33/240-520, 3-156, 32-226, 142-617, W151, W236, W700; S289), each closer with its own qualifiers (S303), and an optional tablet suffix (W400; S24, S29).

Statistics that survive both Markov chains and replicate out of sample (S-DARK-54; held-out sites and IM77-only texts): - W2 once per text: 3 texts with two W2 against 34 expected under a first-order chain (26 under a second-order chain); 3.5 against 47 per 1,000 texts with W2. - Closer paradigm: 9 mutually exclusive closers against a null maximum of 8 after the fragile members W617, W700 and W388 are dropped (S-DARK-41); S289's count of 10 against a shuffled maximum of 5; IM77 replication with a prior prediction, $P = 1.4 \times 10^{-7}$ (S291). - Closer mutual exclusion 0.042 against 0.164; texts with two closers 0.023 against 0.069; the jar never doubled. - No sign repeated within a text: 0.038 against 0.122. - Non-adjacent fixed order of sign pairs: 0.48 against 0.16 (held-out sites 0.65 against 0.03). Held-out pair directions agree with the Mohenjo-daro + Harappa order at 0.89 against 0.80 for a chain (S-DARK-41). - Closer-last on held-out sites 0.42 against a shuffled maximum of 0.21; on identical-reading objects opener-first 0.28, connective-initial 0.003, closer-last 0.76 (S-DARK-41).

Statistics explained by the chain (true, but not evidence of structure beyond adjacency): opener-first (a start-state fact), connective never initial, every 'followed by the jar / preceded by a numeral' gloss, the fish qualifier order (reversed share 0.15 against 0.18-0.20 under the chain), the closers' qualifier specificity (0.09 against 0.08), and the S311 name-element positions (S-DARK-41, S-DARK-54).

Age and uniformity. Across nine chronology series the earliest well-sampled seals show the frame at its all-period rates: opener-first 0.26 against 0.24 overall, closer-last 0.68 against 0.67, W2 in slot two 0.69, repeat rate 0.04, order agreement 0.97; 0-2 of 90 trend arrows for inventory, closer menu and repeat ban are significant; a simulation shows that a system 70-80% formed at the earliest phase would have been detected in 26-40 of 40 runs, so growth above about 10% is excluded. Pre-Mature Kot Diji pottery marks at Harappa (39 tokens) are numerals, W700 and the jar, with W700 and tall-1 over-represented ($P = 0.002$ and 0.001): the counting layer is older than the seal frame (S-DARK-43). On seals, Mohenjo-daro against Harappa at $n = 200$ gives every divergence component within $\times 0.96$ -1.20 of two random halves of one city, none significant (unigram $\times 1.07$, bigram $\times 1.01$, middle inventory $\times 1.01$), and a seal text is shared between the cities more often than between two

halves of one city (2.7% against 2.0%); the held-out sites sit inside the Mohenjo-daro-Harappa band on 32 of 32 band tests at seq_raw and 31 of 32 at the other levels (S-DARK-48.1, 48.3; transcription-robust in IM77). Through the same code Ur III legends at $n = 200$ give a between-site unigram divergence $\times 1.53$ of the within-site value and Linear B syllables $\times 2.64$, with whole-text reuse $22\times$ more local in Ur III (data/derived/dark/loop48_c2.log; row pending).

Generative adequacy. A slot grammar run as a generator fails 36 of 50 held-back statistics; adding site-local whole-text copying, closer dependence on the last middle sign and the object-type effect brings it to the battery's floor (23 of 50, floor 19). What no local mechanism reproduces is long-range attraction and avoidance between signs (127 against 61 attraction pairs under the best generator), the repeat ban and the tight length distribution (S366). Scored as the probability of the next sign, the hand grammar performs exactly like a second-order chain (5.44 against 5.37 bits), and the residual contains no new rule (S-DARK-59.2).

3.2 The designation slot (S-DARK-45, 53, 56, 58, 59)

Withdrawal. The 2 October draft read the texts as graded credentials on two pre-registered predictions (S347–S349): short texts nest whole inside longer ones at $4.1\times$ the within-text-shuffled rate, and 40% of held-out texts reuse a ≥ 3 -sign run from Mohenjo-daro or Harappa ($5\times$). Both pass against a shuffle and fail against a fitted chain: whole-corpus nesting 0.152 against 0.152 under a first-order chain (interval 0.133–0.163); held-out nesting 0.136 against 0.185 under a Mohenjo-daro + Harappa chain ($0.7\times$); held-out run reuse $1.3\times$ against a chain fitted on the held-out sites and $0.7\times$ against a big-city chain. Nesting is reversed on the IM77-only texts (0.058 against 0.19). The grades of a credential were therefore never shown; what remains of the model is the frame of §3.1 and the whole-text copying of §3.1's generator. The 'credential model' entry is grade C (S-DARK-54), and the 'same holder, different grade' reading has no find-spot support (seals sharing a middle but differing in closer: same block 0 of 13–16, same room 0 of 6–9; S-DARK-29).

What the slot is. We describe the middle neutrally as an individual designation written logographically. The evidence: - *Information.* Under the best predictive model (frame automaton, frozen pairs, whole-text caches and a Kneser-Ney chain, fitted on the two big cities), a held-out seal costs 27 bits; 18.7 (69%) are in its 2–3 middle signs at 7.1–7.5 bits each, 4.3 in the frame, 1.5 in the qualifiers and 2.7 in the counts. By slot, the marker is 89% predictable, the suffix 57%, the text end 38–46%, the opener 34%, the closer 32%, the middle 8% (19% with site-local text caches). Through the identical code an Ur III legend gives titles 26–48% predictable and names –18% to 34%, with names 65–71% of the bits; Linear B words 50%. The Indus split matches a name-plus-title legend (S-DARK-59). - *The minimal text.* Of 57 clean one-sign seals, 86% carry a middle-inventory or otherwise unseen sign (16 unseen against 1.0 expected, $P < 0.001$), 4 carry a closer (7%), 1 an opener, 0 a title; two-sign seals are the frame in miniature (135 closers against 70–75 expected, 93% in frame order). Held-out sites 10 of 12, IM77 24 of 27. With the same position-class code, Indus one-sign seals match Latin one-word and Ur III two-word legends, not Linear B ideogram entries (S-DARK-58). - *One pool.* The non-adjacent middle elements are not partitioned by closer or by city: 90% of well-attested elements occur under two or more closers and 97% at two or more sites; once the adjacent qualifier is held, the rest of the middle carries +1.6–2.3% of the closer's information (held-out +0.3–0.5%); co-clustering purity 0.55 against 0.56 under the null, where Ur III name syllables $\times$ profession give 0.55 against 0.37 (S-DARK-45). - *Register, not stock.* Deduplicated middles have uniqueness 0.81, Gini 0.18 and Heaps 0.94, between a uniform ID generator and the Linear B personnel lists and far from Ur III name stocks (0.51 / 0.42 / 0.81); middles of three or more elements are

as distinct as random strings; Chao1 lower bound 12,800–16,400 distinct middles (S-DARK-53).

Relation to personal names: open. Two arguments pull in opposite directions. - *Against phonetically written names.* The contrast used in the 2 October draft (Indus middles 84.6% unique, ratio 0.97 to a bigram model, against Ur III names 0.62; S321–S322) is retired: it measured homonymy in a seal-user population. Ur III names deduplicated to one per person match Indus middles to three decimals (type-token ratio 0.108 against 0.107, Gini 0.72 against 0.73, Zipf slope -1.26 against -1.26), and so do frequency-matched random strings. What separates Indus middles from every one-per-person name list is the absence of a closed edge set: the ten commonest initial and final elements cover 0.24 and 0.27 of middles (random strings 0.26 / 0.26; IM77 0.27 / 0.34), against Ur III initials 0.48, Old Babylonian 0.40 / 0.39, Linear B finals 0.63, Latin initials 0.29; the Indus edge-slot entropy equals that of random strings (0.87). Positional habits exist and transfer to held-out sites (0.333 against 0.258, $z = 5.6$), but with the smallest excess of any system tested (1.29 against Linear B 1.53, Ur III 1.86, Proto-Elamite 1.46). A phonetically spelled name stock has heads or tails (theophoric elements, praenomina, case endings); the Indus middle has none (S-DARK-56). - *For a name-plus-title legend.* The information split (§3.2, first bullet), the minimal-text order (designation kept, closer restored first, opener last) and the office-blind, city-blind pool are the behaviour of names in Ur III legends, and the per-token unpredictability of the middle (7.1–7.5 bits) sits between Sumerian name words (8.7) and name syllables (4.9) (S-DARK-58, S-DARK-59). These two results are compatible if the designation is written logographically (one sign per element, no phonetic heads or tails); the comparison with logographically written given names (Chinese, Japanese, Korean, Vietnamese; S-DARK-63) bears this out: at matched length (2–4 signs) Indus middles are indistinguishable from them on slot binding (0.18 against 0.17–0.27), pair structure, edge openness and frequency shape, so the missing edge set rules out phonetically written names only, not names as such. Two statistics still separate them: 10–12% of Indus middles recur inside longer middles, about twice the shuffle rate and above every logographic list, and the Indus closer depends weakly on the middle (+3–5%), whereas a surname is independent of the given name. 'Middle = name' stays grade C, but no test so far excludes a logographically written personal designation. Would revive a phonetic-name reading: a recurring edge set covering ≥ 0.35 of initials or finals on new material (S-DARK-56).

Inventory. After every defensible merge (shape or reader agreement combined with contextual or free-interchange evidence; 44 variant forms removed from 713 Wells numbers) about 670 signs are in use, about 900 in all (Chao1 913, 95% CI 830–997, or 650–800 after a bias correction calibrated on Linear B, Ur III and Proto-Elamite). A syllabary (60–120 signs) is excluded: even the indefensible single-witness merge leaves 288 signs and destroys the frame. The hapax spectrum ($f1/S$ 0.35) is that of an open logographic inventory of Proto-Elamite or early cuneiform size (S-DARK-52).

3.3 Nameless quantity seals and the Umma sealing (S335–S346)

- **Quantity seals.** Thirty-one texts consist only of 'opener · W2 · N · item', mostly with the item W390 (the tree or branch sign). They come from 8 sites and include two Kalibangan sealings, so the seals were used. Within Mohenjo-daro they are not clustered in one quarter, and seal size does not rise with N ($\rho = -0.38$, $p = 0.14$). W2 here is the opener's marker, not the numeral two (S199, GRAMMAR.md).
- **Counts.** On seals, the numbers attached to the tree sign peak at 3 and 4 (W390 after a numeral 76 of 239 tokens against 33 in shuffles, $z = 9.6$; counts 3×23 , 4×22 , 6×9 , 5×6 ; S350). The Harappa voucher counts (tall 2–4 + W700) peak at the same values. After

'N tree' nothing follows: it fills the closer slot (82 of 95 texts end on the tree, 1 on the jar against 32 expected; S351).

- **The Umma sealing.** The sealing (Ashmolean Museum, Oxford, 1931.120; published by Scheil 1925; Parpola 1994 no. 25) reads 127-705 · 2 · 4 · 390. Its ending, '2-4-390', is the complete text of seven home seals (Mohenjo-daro M-103, M-278, M-984, M-1750, M-1844; Harappa H-2003 and H-55; Dholavira; Surkotada). We are not aware of this equivalence being noted before. Scheil read the strokes after the leftmost sign as three; the museum photograph cannot settle 3 against 4 (S362).
- **Candidate anchor.** In CDLI texts about Meluhha, the most frequently named good is a wood, {gesz}ab-ba me-luh-ha, used for furniture at Umma (P107404, P249043), Ur, Girsu and Isin. We therefore propose W390 = a timber unit. This is a hypothesis with a stated test: sealing backs (log or rope impressions), and Mesopotamian wood accounts with counts of 3-6 units. A point against it: the back of the Umma bulla carries the impression of textile, so it was attached to a cloth-covered bale (Parpola 1994; Scheil 1925; Gadd 1932), which fits bundled goods better than logs. The find-spot 'Umma' rests only on a dealer's report that Scheil himself doubted (Scheil 1925), so the object is best rated unprovenanced; the bulla carries no cuneiform (S353, S362). The tree family's status as the script's main truly counted item is structural (S350) and does not depend on the timber reading.

3.4 Where sealed goods went (S324-S330)

- **All matches are cross-site, but mostly by seal-count.** Eighteen sealings match a seal text exactly. All 18 were found at another site: Mohenjo-daro → Lothal 13, and → Harappa, Kalibangan, Hulas and Rupar; one match runs Harappa → Lothal. A null that permutes seal site labels (keeping each site's seal count) already predicts 93% cross-site matches and 8.8 Mohenjo-daro → Lothal (observed 13, $p = 0.23$), because Mohenjo-daro holds 63% of the long seals. The residual is that 8 of 9 distinct matched texts come from Mohenjo-daro against 56% expected ($p = 0.012$, uncorrected). The direction of flow is therefore suggestive, not established. Match rates are first a function of text length (one-sign seals 4-19%, four or more signs < 0.6%; S-DARK-58.2).
- **Lothal.** The ten Lothal bale sealings (L-161-170) carry 817-2-48-740 with an **elephant**. The only seal with this text, Mohenjo-daro M-895, has a **unicorn**, and Lothal has no elephant seals at all (0 of 43). So the same text was held under different emblems, and the bales were sealed away from Lothal.
- **Kish.** A second Kish seal reproduces the full text of Mohenjo-daro seal M-94 (3-220-590-390-740), preceded by a unit, 416-840, that also occurs at Mohenjo-daro, Dholavira and Harappa. An identical Kish-Mohenjo-daro pair (Mackay 1925 / M-228) has been noted before; this is a second such case.
- **Harappa tablets.** Harappa tablets reproduce seal texts from Mohenjo-daro, Lothal and Dholavira, e.g. 415-220-520, which occurs on 15 Harappa tablets and 1 Mohenjo-daro seal (S344). Whole texts are otherwise copied within a city, not between cities, and the copying is the main thing a slot grammar misses (S366; S-DARK-48.1 reuse $\times 5.6$ when object types are pooled, $\times 0.75$ on seals alone).

4. Discussion

What is established is the frame: five opener units with a once-per-text marker, a closed closer paradigm with mutual exclusion, a tablet suffix, a ban on repeating a sign and a partial order that holds between non-adjacent signs, all beyond a fitted second-order chain and all replicated on held-out sites (S-DARK-54). The frame was complete when the dated

record begins and did not change across the Mature period (S-DARK-43), and it is the same frame, with the same sign and element inventories, in every city (S-DARK-48). The emblem is independent of the text: emblem $\times$ closer association $P = 0.38$ (S329); the same text appears under different animals (S325).

What is withdrawn is the reading of the texts as graded credentials. Nesting and run reuse are what a bigram chain of the same texts produces (S-DARK-41); the office-and-grade picture of the 2 October draft had no evidence beyond adjacency, and the find-spots do not pair seals that would belong to one holder at two grades (S-DARK-29). The frame is compatible with Mukhopadhyay's (2023) administrative reading and with an owner legend; it does not decide between them.

What is open is the referent of the designation slot. It is where the information is (69% of a seal's bits), it is what a one-sign seal keeps, and its elements are drawn from one pan-Indus pool under every closer (S-DARK-45, 58, 59). On the information budget and the minimal-text order it behaves as the name does in an Ur III name-plus-title legend; on its internal make-up it lacks the closed heads and tails that every phonetically written name list has, and it is as open at both edges as a random string (S-DARK-56). The reading consistent with both is a designation written one sign per element, logographically; whether such strings are names, patronymic strings or descriptive epithets is not decided by anything in this paper, and the comparison with logographically written names (loop 63) is pending.

On decipherment. No sound value is recoverable from the distribution of the signs: statistical key-fitting 'reads' planted non-linguistic controls as readily as the real texts (S107, S113-S137), and an annealed sign-to-attribute search against every outside fact (object type, emblem, region, size, material, period) does not beat its permuted-fact control out of sample (S367). Raghavendra (2026) built an emblem system with meaning but no sound values that matched the Indus signs on every reproducible published test. Positive features count more than entropy statistics: counted goods with a self-closing count (S350-S351), a closed set of closers, the same formulas sealed across cities, one-sign labels on pots, no long text on any object (mean 3.3 signs; maximum 17; S354). These show a logographic notation of about 670 signs (S-DARK-52) with a slot grammar; they do not show that the designation slot is or is not a spelled name, and the inventory size and hapax spectrum are those of an open logographic or logosyllabic script, not of a small closed code such as khipu (S-DARK-52). Two cautions. This does not show that the Harappans lacked spoken language, or that they never wrote on perishable materials that have not survived. And the claim is falsifiable: one long running Indus text, of the kind found in every literate neighbour, would overturn the 'no running text' statement, and a recurring edge set on new material would revive the phonetic-name reading. The non-linguistic view was first argued by Farmer, Sproat & Witzel (2004) and disputed by Rao et al. (2009); the tests here add chain-calibrated and held-out evidence to that debate, and they move the question from 'language or not' to 'what does the designation slot denote'.

5. Limitations

- The corpus is a small sample of all seals ever made: Chao1 estimates about 900-950 signs, of which about 240 are unseen (S309, S-DARK-52), and 12,800-16,400 distinct middles against 1,971 seen (S-DARK-53). Most texts come from Mohenjo-daro and Harappa (76%); the dated record rests on those two cities and the pre-Mature evidence on Harappa alone (S-DARK-43).
- Wells and IM77 are about 70% the same objects (S312). Agreement between them is transcription-robust, not replication; the independent evidence is the held-out Wells sites and the 324 IM77-only texts, which have power only for the closer system (S-DARK-54).

- Sign identities depend on the transcription. Allograph merges were run at three levels and every result above holds at each; W617 and W700 as closers, and any count that separates the jar W740 from the marked jar W741, are transcription-fragile (S-DARK-24).
- Nulls that keep text lengths cannot end a text, so finality claims were tested against an END-state chain; opener-first is a start-state fact in every chain (S-DARK-54).
- Several frame rules are shown only at Mohenjo-daro + Harappa and are inside wide bands elsewhere for lack of power: opener/W400 exclusion, numeral exclusivity, marker-second, long-range attraction (S-DARK-54).
- The name comparison mixes syllabic and logographic scripts. One-per-person deduplication removes the homonymy artefact (S-DARK-56); logographic given names (S-DARK-63) match Indus middles on most statistics; the substring result rests mainly on the Japanese, Vietnamese and ancient Chinese lists, since modern Chinese and Korean given names are almost all two signs long.
- merged-corpus-canonical.json was built from an older inscriptions.csv (S-DARK-23); the rebuild moves no statistic (S-DARK-41.1), but whole-corpus counts should be re-quoted from the rebuilt file before publication.
- The timber reading of W390 is a hypothesis, not a demonstration; the textile impression on the back of the Umma bulla argues against it, and the Umma provenance is a dealer's report (S353, S362).
- A literature check (web; Mukhopadhyay 2023; Yadav et al.; Laursen 2010; arXiv 2608.02999) found no prior report of results 3.3 or 3.4 (cross-site statistics). A specialist check of Parpola's CISI commentary is still pending.

6. Reproducibility

The scripts are in tools/. Each is named after its STRATEGIES row: - dark_loop41_c1.py-dark_loop41_c4.py, dark_loop41_common.py (S-DARK-41, red team) - dark_loop54_common.py, dark_loop54_claims.py, dark_loop54_c1.py-dark_loop54_c4.py (S-DARK-54; grades in data/derived/dark/loop54_regrade.csv) - dark_loop43.py (S-DARK-43), dark_loop45.py (S-DARK-45), dark_loop52_c1.py-dark_loop52_c3b.py (S-DARK-52), dark_loop53.py (S-DARK-53), dark_loop56_prep.py, dark_loop56.py (S-DARK-56), dark_loop58.py, dark_loop58_calib.py (S-DARK-58), dark_loop59_common.py, dark_loop59_c1.py-dark_loop59_c3.py (S-DARK-59) - strat_adequacy.py (S366), strat_anneal.py (S367) - strat_credential.py, strat_credential_heldout.py (S347, S349; superseded by S-DARK-41) - strat_namecalib.py, strat_namecalib2.py (S321-S322; retired by S-DARK-56) - strat_batch1.py (S336-S338), strat_sealflow.py (S324), strat_branches.py, strat_590title.py, strat_abroadsuffix.py (S326-S328) - parse_all.py (S310), strat_potcompare.py (S354, S355), strat_counted.py, strat_countedcloser.py (S350, S351) - build_paper_pdf.py (this document to docs/Garg_2026_Indus_seals_credentials.pdf)

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